

OBJECT_AVOIDER

About :

Pluto drone is equipped with a IR sensor to detect external objects while maneuring in the air and avoid crashing into them

Add-On used :

A IR sensor module consists of a IR led and receiver which detects the reflection of IR rays from nearby objects. It returns 0 if any subject is bought very near to the sensor, otherwise 1.

Components and Connection:

Pluto drone, IR_sensor module and connecting wires.

Connect the Sound sensor on the RC port of Primus V5/X2 board

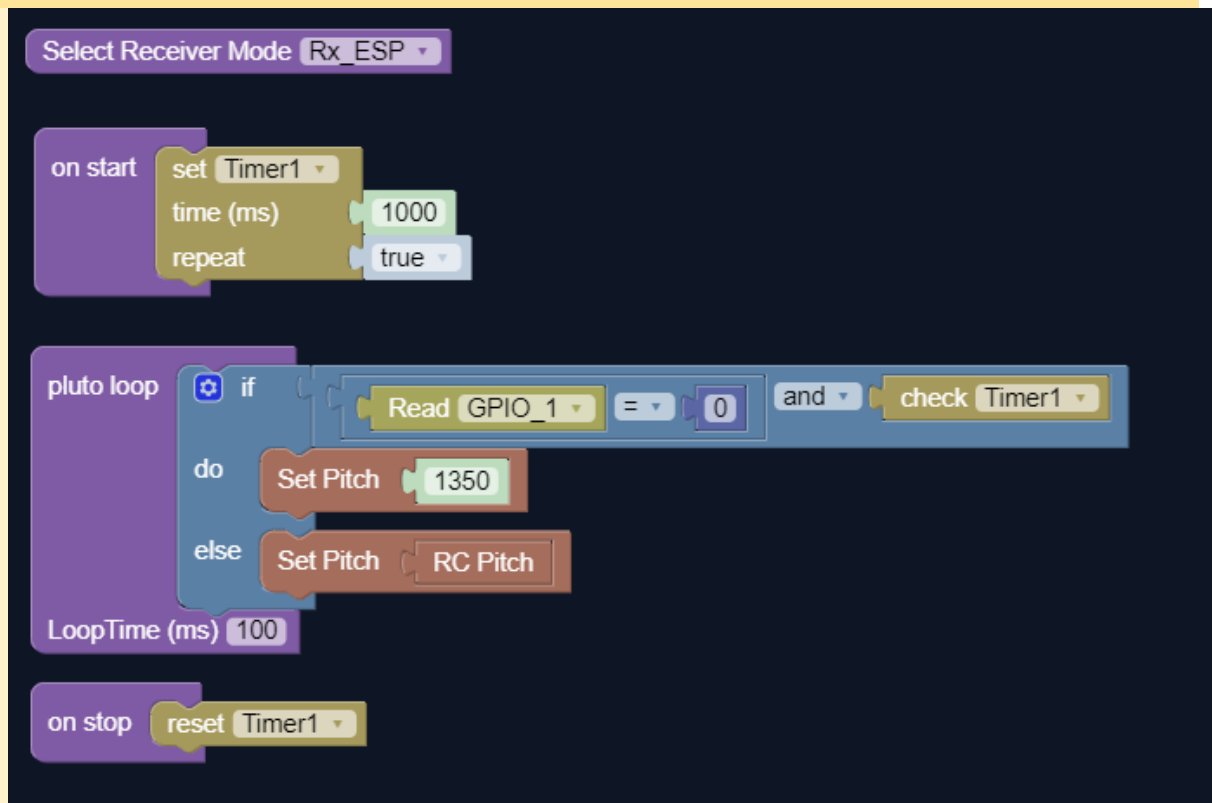
Primus V5/X2 pin	Peripherals
VCC on RC port	VCC on IR_module
GND on RC port	GND on IR_module
Signal on RC port	D0 on IR_module

How does it work?

Fame sensor module is basically a proximity sensor to detect objects nearby drone:

- Normal reading → 1 = normal operation
- Object detected → 0 = Move in opposite direction

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Working:

Initially, we set Receiver mode as ESP and set a timer. When loop begins, we constantly evaluate whether there is any object coming in close proximity to our drone. If it is it provides thrust in opposite direction for amount of time defined in timer.

Moving further

Try adding more features to make a Channel/tunnel follower drone that will move on its own, without colliding.
Try making a wall following drone.

